

Functional Requirements Specification (FRS)

AI Trust & Safety Auditing Agent for AI-Based AIS

Project: Project 1 of 3
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1. Introduction

1.1 Purpose

This Functional Requirements Specification (FRS) defines the implementation-ready requirements for the AI Trust & Safety Auditing Agent, a Council of Experts system that evaluates AI-based Accounting Information Systems (AIS) across five dimensions: Governance, Fairness, Security/Privacy, Explainability, and Accuracy.

1.2 Document Scope

This FRS covers the requirements for Projects 2 and 3 to implement and integrate the auditing agent based on the foundation established in Project 1: - Research Synthesis (Objective 1) - Machine-Readable Schemas (Objective 2) - System Architecture (Objective 3)

1.3 Intended Audience

- **Projects 2 & 3 Development Teams:** Implementation guidance
- **Dr. Andrés Fortino (Sponsor):** Requirements approval and validation
- **NYU SPS Faculty:** Capstone evaluation
- **Future Maintenance Teams:** System understanding and enhancement

1.4 Document Conventions

Requirement Priority Levels: - **MUST (M):** Critical requirement; system cannot function without it - **SHOULD (S):** Important requirement; significantly impacts value - **COULD (C):** Desirable requirement; nice-to-have if resources permit

Requirement IDs: Format: FR- [Category] - [Number] (e.g., FR-INPUT-001)

2. Scope and Objectives

2.1 In Scope for Projects 2 & 3

Project 2 (Implementation): - Five module implementations (Governance, Fairness, Security, Explainability, Accuracy) - ChatGPT API integration for criterion evaluation - Input parser and validation - Basic web interface (Streamlit or FastAPI) - SQLite database for audit logs - Proof of Concept → MVP transition

Project 3 (Integration): - Synthesis layer with 10 interdependency triggers - Divergence detection and HITL escalation - Report generation (PDF, JSON, CSV outputs) - End-to-end testing with sample AI systems - Deployment to NYU AI Sandbox

2.2 Out of Scope

- Trial execution with real companies (future work beyond capstone)
- Expert panel validation (documented in Technology Trial Plan but not implemented)
- Production deployment outside NYU Sandbox
- Multi-user authentication and role-based access control (future enhancement)
- Real-time continuous monitoring (future enhancement)

2.3 Objectives Alignment

This FRS supports completion of **Objective 4** from Project 1 proposal:

Objective 4: Deliver an implementation-ready requirements package and a lightweight PoC to support downstream build.

Metric: Sponsor acceptance of FRS feasibility and clarity; PoC runs successfully and produces consistent scorecard/report outputs for the sample scenarios.

3. Functional Requirements

3.1 Input Processing Requirements

FR-INPUT-001 (M): Evidence Artifact Upload - Description: System shall accept evidence artifacts in multiple formats (PDF, DOCX, CSV, JSON, PNG, JPEG) via web interface. - **Acceptance Criteria:** - Upload supports drag-and-drop and file browser selection - File size limit: 50MB per artifact, 500MB total per audit - Supported formats validated on upload (reject unsupported types) - Evidence stored with timestamp and original filename preserved - **Priority:** MUST - **Maps to Objective:** Objective 2 (Schema defines evidence artifact structure)

FR-INPUT-002 (M): AI System Metadata Collection - Description: System shall collect AI system metadata via web form. - **Acceptance Criteria:** - Form captures: system name, version, deployment date, business process, geographic scope - Geographic scope options: US, EU, Other (multi-select) - Special category data flag (Yes/No) - All required fields validated before submission - **Priority:** MUST - **Maps to Objective:** Objective 3 (Architecture defines input schema)

FR-INPUT-003 (M): JSON Schema Validation - Description: System shall validate all inputs against JSON schemas from Objective 2. - **Acceptance Criteria:** - Input parser validates metadata against schema - Evidence artifacts mapped to criterion IDs (G1.1 - A5.5) - Validation errors display user-friendly messages - Invalid inputs rejected with specific error explanations - **Priority:** MUST - **Maps to Objective:** Objective 2 (28 criterion schemas created)

FR-INPUT-004 (S): Geographic Exemption Handling - Description: System shall automatically apply N/A to criteria not applicable based on geographic scope. - **Acceptance Criteria:** - If geographic_scope excludes "EU", mark GDPR-only criteria as N/A - If geographic_scope excludes "US", mark CCPA-only criteria as N/A - N/A criteria excluded from score calculations (weight renormalization) - Exemption logic documented in audit log - **Priority:** SHOULD - **Maps to Objective:** Objective 2 (Schema includes exemption_status field)

3.2 Module Evaluation Requirements

FR-MODULE-001 (M): Five Independent Module Implementations - Description: System shall implement five expert modules, each evaluating its assigned criteria independently. - **Acceptance Criteria:** - Module 1 (Governance): Evaluates G1.1 - G1.7 (7 criteria) - Module 2 (Fairness): Evaluates F2.1 - F2.4 (4 criteria) - Module 3 (Security): Evaluates S3.1 - S3.6 (6 criteria) - Module 4 (Explainability): Evaluates E4.1 - E4.5 (5 criteria) - Module 5 (Accuracy): Evaluates A5.1 - A5.5 (5 criteria) - Each module implements BaseModule interface (Objective 3) - Modules have zero side effects (stateless execution) - **Priority:** MUST - **Maps to Objective:** Objectives 2 & 3 (Schemas + Architecture)

FR-MODULE-002 (M): ChatGPT Integration for Criterion Evaluation - Description: System shall use ChatGPT (GPT-4) to analyze evidence and score each criterion. -

Acceptance Criteria: - ChatGPT API integration via OpenAI Python SDK - JSON mode enabled for structured output - Temperature set to 0.2 for consistency - Each criterion receives score 0.0-1.0 and text justification - API errors handled gracefully (retry with exponential backoff) - **Priority:** MUST - **Maps to Objective:** Objective 3 (Tech Stack specifies ChatGPT as main LLM)

FR-MODULE-003 (M): Criterion Scoring and Classification - **Description:** System shall score each criterion 0.0-1.0 and classify per SOX 404 standards. - **Acceptance Criteria:** - Score range enforced: $0.0 \leq \text{score} \leq 1.0$ - Classification logic: - Pass: $\text{score} > 0.75$ - Significant Deficiency: $0.51 \leq \text{score} \leq 0.75$ - Material Weakness: $\text{score} \leq 0.5$ - N/A: criterion not applicable - Justification text required for all scores - Evidence references linked to scores - **Priority:** MUST - **Maps to Objective:** Objective 2 (Schema defines scoring scale)

FR-MODULE-004 (S): Module-Level Score Aggregation - **Description:** System shall compute module-level aggregate scores as weighted average of criterion scores. - **Acceptance Criteria:** - Weighted average calculated (all criteria weighted equally within module) - N/A criteria excluded from calculation - Module score range: 0.0-1.0 - Module classification based on aggregate score (same thresholds as criteria) - **Priority:** SHOULD - **Maps to Objective:** Objective 3 (Architecture defines aggregation logic)

FR-MODULE-005 (S): Regulatory Citation Mapping - **Description:** System shall map criterion scores to relevant regulatory citations (GDPR, CCPA, SOX). - **Acceptance Criteria:** - Each GDPR/CCPA criterion includes applicable article/section references - Citations displayed in module output - Citations carried through to final report - Citation accuracy validated against Schema Documentation v3.1 - **Priority:** SHOULD - **Maps to Objective:** Objective 2 (Schema includes regulatory_citations field)

3.3 Synthesis Layer Requirements

FR-SYNTH-001 (M): Interdependency Trigger Engine - **Description:** System shall implement 10 interdependency triggers that coordinate findings across modules. - **Acceptance Criteria:** - All 10 triggers from Objective 3 Architecture implemented: 1. Governance gap → Fairness escalation 2. Governance weakness → Security escalation 3. ROPA incomplete → Privacy risk flag 4. Explainability failure → Global confidence flag 5. Fairness failure → Accuracy re-evaluation 6. Drift detection → Adversarial review 7. Breach failure → All privacy criteria escalate 8. Audit trail gaps → Evidence reliability flag 9. Opt-out missing → Transparency escalation 10. Materiality adjustment (GDPR/CCPA fines) - Trigger activation logged with source and affected criteria - Flags applied to affected findings (“Enhanced Testing”, “Higher Risk”, “Lower Confidence”) - **Priority:** MUST - **Maps to Objective:** Objective 3 (Architecture defines 10 triggers)

FR-SYNTH-002 (M): Divergence Detection - **Description:** System shall detect score divergences ≥ 20 points between modules evaluating overlapping criteria. - **Acceptance Criteria:** - Divergence calculated: $|\text{Score_ModuleA} - \text{Score_ModuleB}|$ - Threshold: Divergence ≥ 0.20 (20 percentage points) - Divergent findings flagged for review - Both module scores and evidence preserved for comparison - **Priority:** MUST - **Maps to Objective:** Objective 3 (Three-stage synthesis includes divergence handling)

FR-SYNTH-003 (S): Human-in-the-Loop (HITL) Escalation - Description: System shall escalate divergent findings (≥ 20 points) to human review. - **Acceptance Criteria:** - HITL escalation queue created for divergent findings - Escalation presents both module outputs side-by-side - Evidence used by each module displayed - Adjudication interface allows human to set final score - Human decision logged with timestamp and username - **Priority:** SHOULD - **Maps to Objective:** Objective 3 (Architecture includes HITL escalation)

FR-SYNTH-004 (M): Overall Audit Score Calculation - Description: System shall calculate overall audit score as weighted average of module scores. - **Acceptance Criteria:** - All 5 module scores averaged (equal weights for MVP) - N/A modules excluded from calculation - Overall score range: 0.0-1.0 - Overall classification: Pass, Significant Deficiency, or Material Weakness - **Priority:** MUST - **Maps to Objective:** Objective 3 (Synthesis output schema)

FR-SYNTH-005 (M): Materiality Calculation - Description: System shall calculate financial materiality including GDPR/CCPA exposure. - **Acceptance Criteria:** - Base materiality threshold accepted as input (USD) - GDPR fine calculation: €20M or 4% annual turnover (whichever higher) - CCPA private right of action: \$100-\$750 per consumer - Data breach costs estimated (optional: \$150 per record) - Total exposure calculated and compared to materiality threshold - Material Weaknesses flagged if exposure > threshold - **Priority:** MUST - **Maps to Objective:** Objective 3 (Interdependency trigger 10: materiality adjustment)

3.4 Report Generation Requirements

FR-REPORT-001 (M): Audit Scorecard Generation - Description: System shall generate comprehensive audit scorecard with all scores and classifications. - **Acceptance Criteria:** - Scorecard includes: - Overall audit score and classification - 5 module scores and classifications - 28 criterion scores and classifications - N/A criteria clearly marked - Timestamp and auditor information - Format: JSON (machine-readable) and PDF (human-readable) - **Priority:** MUST - **Maps to Objective:** Objective 4 (PoC includes scorecard generator)

FR-REPORT-002 (M): Material Weakness Report - Description: System shall generate Material Weakness report listing all Material Weaknesses with remediation guidance. - **Acceptance Criteria:** - Report includes all criteria with score ≤ 0.5 - For each Material Weakness: - Criterion ID and name - Score and justification - Materiality impact (USD) - Regulatory exposure (GDPR/CCPA citations) - Remediation priority (Critical, High, Medium) - Recommended actions - Report sortable by priority - **Priority:** MUST - **Maps to Objective:** Objective 3 (Synthesis output includes material_weaknesses)

FR-REPORT-003 (S): Evidence Traceability Log - Description: System shall generate evidence traceability log mapping scores to evidence artifacts. - **Acceptance Criteria:** - Log format: CSV - Columns: Criterion ID, Score, Classification, Evidence Artifact Names, Upload Timestamps - Enables audit trail reconstruction - Supports compliance verification - **Priority:** SHOULD - **Maps to Objective:** Objective 3 (Architecture includes evidence_references)

FR-REPORT-004 (S): Interdependency Trigger Log - Description: System shall generate log of all interdependency triggers activated during audit. - **Acceptance Criteria:** - Log includes: - Trigger ID and description - Source criterion that activated trigger - Affected criteria - Action taken (escalation/flag) - Timestamp - Format: JSON and PDF - **Priority:** SHOULD - **Maps to Objective:** Objective 3 (Synthesis output includes interdependency_triggers_activated)

FR-REPORT-005 (C): HITL Escalation Summary - Description: System shall generate summary of all HITL escalations and adjudications. - **Acceptance Criteria:** - Summary includes: - Criterion ID - Divergence score - Module scores (both) - Human decision and final score (if adjudicated) - Adjudicator name and timestamp - Status: Pending or Adjudicated - **Priority:** COULD - **Maps to Objective:** Objective 3 (Synthesis output includes hitl_escalations)

3.5 Data Management Requirements

FR-DATA-001 (M): Audit Log Persistence - Description: System shall persist all audit data to SQLite database for reproducibility. - **Acceptance Criteria:** - Database tables: Audits, Evidence, Scores, Triggers, Escalations - All scores, justifications, and evidence references stored - Audit log immutable (no updates, only inserts) - Timestamp all database entries - **Priority:** MUST - **Maps to Objective:** Objective 3 (Tech Stack specifies SQLite)

FR-DATA-002 (S): Evidence Artifact Storage - Description: System shall store evidence artifacts on filesystem with secure naming. - **Acceptance Criteria:** - Storage location: /evidence/{audit_id}/{criterion_id}/ - Filenames: UUID + original extension (prevents collisions) - Original filename preserved in metadata - Storage limit enforced (500MB per audit) - **Priority:** SHOULD - **Maps to Objective:** Objective 3 (Deployment architecture)

FR-DATA-003 (S): Audit History Query - Description: System shall support querying audit history by date, system, or classification. - **Acceptance Criteria:** - Query filters: Date range, AI system name, overall classification - Results sortable by date (newest first) - Query results exportable to CSV - Pagination for large result sets (>100 audits) - **Priority:** SHOULD - **Maps to Objective:** Future enhancement (noted in Tech Stack roadmap)

3.6 User Interface Requirements

FR-UI-001 (M): Web-Based Audit Workflow - Description: System shall provide web interface for complete audit workflow. - **Acceptance Criteria:** - Pages/screens: 1. New Audit (metadata entry + evidence upload) 2. Audit Progress (real-time module execution status) 3. Audit Results (scorecard, reports) 4. Audit History (past audits) - Navigation between pages - Responsive design (desktop and tablet) - **Priority:** MUST - **Maps to Objective:** Objective 3 (Tech Stack specifies Streamlit or FastAPI)

FR-UI-002 (S): Real-Time Progress Indicator - Description: System shall display real-time progress during audit execution. - **Acceptance Criteria:** - Progress bar showing % complete (0-100%) - Module-level status (Pending, In Progress, Complete) - Estimated

time remaining - WebSocket or polling for updates - **Priority:** SHOULD - **Maps to Objective:** Objective 3 (FastAPI supports WebSocket)

FR-UI-003 (S): Report Download Interface - Description: System shall provide one-click download for all generated reports. - **Acceptance Criteria:** - Download buttons for: Scorecard (PDF, JSON), Material Weakness Report (PDF), Evidence Log (CSV), Trigger Log (PDF) - Downloads preserve original filenames with timestamp - Concurrent downloads supported - **Priority:** SHOULD - **Maps to Objective:** Objective 3 (Output layer in architecture)

FR-UI-004 (C): HITL Adjudication Interface - Description: System shall provide interface for human adjudication of divergent findings. - **Acceptance Criteria:** - Displays both module scores side-by-side - Shows evidence used by each module - Allows human to set final score (0.0-1.0) - Captures justification text - Saves adjudication with timestamp - **Priority:** COULD - **Maps to Objective:** Objective 3 (HITL escalation in synthesis layer)

4. Non-Functional Requirements

4.1 Performance Requirements

NFR-PERF-001 (M): Audit Execution Time - Description: Complete audit shall execute in ≤ 30 minutes for standard evidence volume. - **Acceptance Criteria:** - Standard evidence: 28 documents, ~ 10 pages each - Execution time measured from submission to report generation - 90th percentile ≤ 30 minutes - **Priority:** MUST

NFR-PERF-002 (S): Concurrent Audit Support - Description: System shall support at least 3 concurrent audits without performance degradation. - **Acceptance Criteria:** - 3 audits running simultaneously - Each audit completes within ≤ 35 minutes ($\leq 17\%$ slowdown acceptable) - No errors or timeouts - **Priority:** SHOULD

NFR-PERF-003 (S): API Rate Limit Handling - Description: System shall gracefully handle ChatGPT API rate limits. - **Acceptance Criteria:** - 429 errors trigger exponential backoff (1s, 2s, 4s, 8s, 16s) - Maximum 5 retry attempts - User notified if rate limit persists after 5 retries - **Priority:** SHOULD

4.2 Usability Requirements

NFR-USAB-001 (M): Ease of Evidence Upload - Description: Auditor shall upload 28 evidence sets in ≤ 10 minutes. - **Acceptance Criteria:** - Batch upload supported (multiple files at once) - Drag-and-drop functional - Upload progress indicator - Clear error messages for invalid files - **Priority:** MUST

NFR-USAB-002 (S): Report Readability - Description: Generated reports shall be readable by non-technical stakeholders. - **Acceptance Criteria:** - Executive summary in plain language - Technical jargon defined in glossary - Visual elements (tables, color coding) used appropriately - Font size ≥ 11 pt for body text - **Priority:** SHOULD

4.3 Reliability Requirements

NFR-RELI-001 (M): Error Handling - Description: System shall handle errors gracefully without data loss. - **Acceptance Criteria:** - All exceptions caught and logged - User-friendly error messages displayed - Partial audit state saved (resume capability) - Database transactions ensure data consistency - **Priority:** MUST

NFR-RELI-002 (S): Audit Reproducibility - Description: Re-running audit with identical evidence shall produce identical scores (± 0.05). - **Acceptance Criteria:** - Same evidence + same ChatGPT prompts = same scores (within tolerance) - All inputs logged for reproducibility - Random seed controlled (if applicable) - **Priority:** SHOULD

4.4 Security Requirements

NFR-SEC-001 (M): API Key Protection - Description: OpenAI API key shall be stored securely (not hardcoded). - **Acceptance Criteria:** - API key loaded from environment variable - No API keys in source code or version control - Key rotation supported (change key without code changes) - **Priority:** MUST

NFR-SEC-002 (S): Evidence Data Privacy - Description: Uploaded evidence shall be accessible only to the auditor who uploaded it (MVP: single user). - **Acceptance Criteria:** - Evidence stored in isolated directories per audit - File permissions restrict access (OS-level) - No public exposure of evidence files - **Priority:** SHOULD

4.5 Maintainability Requirements

NFR-MAINT-001 (M): Code Documentation - Description: All modules shall have docstrings and inline comments. - **Acceptance Criteria:** - All functions/classes have docstrings (description, args, returns) - Complex logic has inline comments - README.md explains setup and usage - **Priority:** MUST

NFR-MAINT-002 (S): Test Coverage - Description: Code shall have $\geq 70\%$ test coverage. - **Acceptance Criteria:** - Unit tests for all modules - Integration tests for end-to-end workflow - pytest coverage report $\geq 70\%$ - **Priority:** SHOULD

5. Acceptance Criteria

5.1 Overall System Acceptance

The AI Trust & Safety Auditing Agent shall be considered **accepted** when:

1. All MUST (M) requirements implemented and tested
2. $\geq 80\%$ of SHOULD (S) requirements implemented
3. All 5 modules produce valid outputs per JSON schema
4. At least 3 sample audits complete successfully
5. Sponsor approves deliverables (FRS + PoC)
6. System deployable to NYU AI Sandbox without errors

5.2 Requirement-Level Acceptance

Each requirement shall be considered **accepted** when:

1. Implementation matches requirement description
2. All acceptance criteria met
3. Automated tests pass (where applicable)
4. Manual testing validates functionality
5. Documentation updated

5.3 PoC Acceptance (Objective 4 Specific)

The Proof of Concept shall be considered **accepted** when:

1. Schema validator successfully validates all 28 criterion schemas
2. Scorecard generator produces valid JSON and PDF outputs
3. Report skeleton generator creates structured audit report template
4. At least 3 sample scenarios run successfully with consistent outputs
5. Sponsor approves PoC as sufficient foundation for Project 2

6. Requirements Traceability Matrix

6.1 Functional Requirements → Objectives Mapping

Req ID	Requirement Name	Obj 1	Obj 2	Obj 3	Obj 4	Priority
FR-INPUT-001	Evidence Upload		✓		✓	M
FR-INPUT-002	Metadata Collection			✓	✓	M
FR-INPUT-003	Schema Validation		✓		✓	M
FR-INPUT-004	Geographic Exemption		✓			S
FR-MODULE-001	Five Modules		✓	✓		M
FR-MODULE-002	ChatGPT Integration	✓		✓		M
FR-MODULE-003	Scoring & Classification		✓			M
FR-MODULE-004	Module Aggregation			✓		S
FR-MODULE-005	Regulatory Citations	✓	✓			S
FR-SYNTH-001	Interdependency Triggers			✓		M
FR-SYNTH-002	Divergence Detection			✓		M
FR-SYNTH-003	HITL Escalation			✓		S
FR-SYNTH-004	Overall Score			✓		M
FR-SYNTH-005	Materiality Calculation	✓		✓		M
FR-REPORT-001	Audit Scorecard				✓	M

Req ID	Requirement Name	Obj 1	Obj 2	Obj 3	Obj 4	Priority
FR-REPORT-002	Material Weakness Report	✓	✓		✓	M
FR-REPORT-003	Evidence Traceability				✓	S
FR-REPORT-004	Trigger Log			✓	✓	S
FR-REPORT-005	HITL Summary			✓		C
FR-DATA-001	Audit Log Persistence			✓		M
FR-DATA-002	Evidence Storage			✓		S
FR-DATA-003	Audit History Query					S
FR-UI-001	Web Workflow			✓		M
FR-UI-002	Progress Indicator			✓		S
FR-UI-003	Report Downloads				✓	S
FR-UI-004	HITL Adjudication			✓		C

Legend: - **Obj 1:** Research Synthesis - **Obj 2:** Machine-Readable Schemas - **Obj 3:** System Architecture - **Obj 4:** Requirements & PoC - **M:** MUST, **S:** SHOULD, **C:** COULD

6.2 Objectives → Requirements Coverage

Objective	Requirements Count	MUST	SHOULD	COULD
Objective 1 (Research)	3	2	1	0
Objective 2 (Schemas)	9	6	2	0
Objective 3 (Architecture)	17	11	4	2
Objective 4 (Req & PoC)	7	4	3	0

Coverage Analysis: - All objectives have associated requirements - Objective 3 (Architecture) has highest requirement count (expected - foundation for implementation) - 23 MUST requirements ensure core functionality - 10 SHOULD requirements enhance value - 2 COULD requirements provide nice-to-have features

7. Constraints and Assumptions

7.1 Technical Constraints

Constraint	Description	Impact	Mitigation
NYU AI Sandbox	Limited to sandbox environment	No production deployment outside NYU	Design for portability (Docker-ready for future)
API Access	Sandbox may restrict external API calls	ChatGPT access uncertain	Verify “API Access” mode enabled; document fallback (Azure OpenAI)

Constraint	Description	Impact	Mitigation
Single User (MVP)	No multi-user authentication	One auditor at a time	Document as future enhancement; sufficient for capstone
SQLite (MVP)	Embedded database	Limited scalability	Migration path to PostgreSQL documented
Python 3.10+	Requires specific Python version	Older environments incompatible	Verify NYU Sandbox Python version

7.2 Assumptions

1. **ChatGPT Reliability:** Assumes ChatGPT API is stable and available during audit execution.
2. **Evidence Quality:** Assumes auditors provide complete and relevant evidence artifacts.
3. **Sponsor Availability:** Assumes Dr. Fortino available for reviews and approvals within 1 week.
4. **NYU Sandbox Access:** Assumes Projects 2 & 3 teams have NYU Sandbox access throughout development.
5. **JSON Schema Stability:** Assumes 28 criterion schemas from Objective 2 remain stable (no major changes).
6. **Project 2 Resources:** Assumes Project 2 team has Python development skills.
7. **Internet Access:** Assumes NYU Sandbox “API Access” mode enabled for ChatGPT calls.

7.3 Dependencies

External Dependencies: - OpenAI ChatGPT API (GPT-4) - Python 3.10+ environment - NYU AI Sandbox infrastructure

Internal Dependencies (Project 1 Deliverables): - Research Synthesis v3.1 (Objective 1) → Informs requirement priorities - 28 JSON Schemas (Objective 2) → Defines data structures - System Architecture (Objective 3) → Defines module interfaces - Proof of Concept (Objective 4) → Validates technical feasibility

8. Implementation Priorities

8.1 Phase 1: Core Functionality (Project 2 - Weeks 1-6)

Sprint 1 (Weeks 1-2): Input Processing - FR-INPUT-001: Evidence Upload - FR-INPUT-002: Metadata Collection - FR-INPUT-003: Schema Validation - FR-DATA-001: Audit Log Persistence

Sprint 2 (Weeks 3-4): Module Implementation - FR-MODULE-001: Implement 5 modules (BaseModule interface) - FR-MODULE-002: ChatGPT Integration - FR-MODULE-003: Scoring & Classification

Sprint 3 (Weeks 5-6): Basic Reporting - FR-REPORT-001: Audit Scorecard - FR-REPORT-002: Material Weakness Report - FR-UI-001: Web Workflow (basic version)

8.2 Phase 2: Enhanced Features (Project 2 - Weeks 7-10)

Sprint 4 (Weeks 7-8): Synthesis Layer - FR-SYNTH-001: Interdependency Triggers (all 10) - FR-SYNTH-004: Overall Score Calculation - FR-SYNTH-005: Materiality Calculation

Sprint 5 (Weeks 9-10): Advanced Reporting - FR-REPORT-003: Evidence Traceability - FR-REPORT-004: Trigger Log - FR-UI-003: Report Downloads

8.3 Phase 3: Integration & Testing (Project 3 - Weeks 11-14)

Sprint 6 (Weeks 11-12): Synthesis Completion - FR-SYNTH-002: Divergence Detection - FR-SYNTH-003: HITL Escalation (if time permits) - FR-UI-002: Progress Indicator

Sprint 7 (Weeks 13-14): Testing & Deployment - End-to-end testing with ≥ 3 sample AI systems - NYU Sandbox deployment - Documentation finalization

8.4 Optional Enhancements (If Time Permits)

- FR-INPUT-004: Geographic Exemption Handling (automated)
- FR-MODULE-004: Module Aggregation (with custom weights)
- FR-UI-004: HITL Adjudication Interface
- FR-REPORT-005: HITL Summary
- FR-DATA-003: Audit History Query

9. Glossary

Term	Definition
AI System	An AI-based Accounting Information System (AIS) being audited
Criterion	One of 28 evaluation criteria (G1.1 - A5.5)
Module	One of 5 expert evaluation modules (Governance, Fairness, Security, Explainability, Accuracy)
Evidence Artifact	Document, data file, or screenshot supporting criterion evaluation
Material Weakness	SOX 404 classification for criterion score ≤ 0.5
Significant Deficiency	SOX 404 classification for criterion score 0.51-0.75
Pass	SOX 404 classification for criterion score > 0.75
Interdependency Trigger	Rule that coordinates findings across modules
HITL	Human-in-the-Loop (manual review for divergent findings)

Term	Definition
Divergence	Score difference ≥ 20 points between modules on overlapping criteria
Synthesis	Three-stage process combining module outputs
PoC	Proof of Concept (schema validator + scorecard/report generators)

Appendices

Appendix A: JSON Schema Files

All 28 criterion schemas available at: /schemas/ (Objective 2 deliverable)

Appendix B: Architecture Diagrams

System architecture diagrams available in: System_Architecture_Tech_Stack.docx (Objective 3 deliverable)

Appendix C: PoC Source Code

Proof of Concept code available at: /poc/ (Objective 4 deliverable - see separate files)

Appendix D: Sample Scenarios

Three sample audit scenarios available at: /poc/samples/ (Objective 4 deliverable)